EECS 563 Midterm 1 – Study Guide – Fall 2026

The midterm will cover material on the introduction, application layer, and transport layer from the course textbook (Kurose and Ross, *Computer Networking: A Top-Down Approach*). The midterm material is mainly from Chapters 1–3, the corresponding lecture slides, and homework.

Chapter 1: Introduction

– What is the Internet?
– Hosts, end systems
– Protocol
– Network edge and Network core
– Access networks: Wired and wireless access networks
– Network core: Circuit switching and packet switching
– Circuit switching: FDMA, TDMA
– Packet switching: Store-and-forward concept
– Packet switching: Statistical multiplexing
– Packet switching vs. Circuit switching: Number of users supported; probability calculations
– The Internet: a network of networks (*internetwork*)
– Delay, loss, throughput in packet-switched networks
– Types of delay: processing, queueing, transmission, propagation
– Average queueing delay vs. traffic intensity
– Throughput in computer networks
– Protocol layers
– Layering as an architectural principle, services provided by each layer, encapsulation/decapsulation
– Layers in OSI, TCP/IP models

Chapter 2: Application Layer

– Principles of network applications
– Possible architectures of applications: Client-server vs. P2P architectures
– Client vs. server
– Communicating processes
– Addressing a process: IP address and port number
– Network application requirements: data integrity, throughput, timing
– Transport services: reliable vs. unreliable data transfer
– TCP vs. UDP services
– Web and HTTP
– Base HTML file and referenced objects
– HTTP request and response concepts
– Persistent and non-persistent HTTP
– HTTP and its underlying transport layer protocol (TCP vs. QUIC)
– HTTP request/response messages (e.g., GET method)
– HTTP cookies
– Web caching and performance improvement
– Conditional GET
– Head of line blocking (HOL) in HTTP
– Email components and protocols: SMTP, POP3, and IMAP
– DNS: DNS architecture, hierarchy of name servers
– Iterative vs. recursive queries
– DNS resource records and messages
– P2P applications
– P2P file distribution vs. client-server
– BitTorrent
– Video streaming and content distribution networks (CDN)
– DASH protocol
– TCP and UDP socket programming

Chapter 3: Transport Layer

– Transport-layer services
– Logical *end-to-end* connection between *processes* vs. end-to-end connection between *hosts*

– Multiplexing and demultiplexing in UDP vs. TCP

– Source & destination IP addresses, port numbers

– Advantages and disadvantages of UDP

– UDP header structure and UDP checksum

– Principles of reliable data transfer (RDT)

– RDT over channels with errors, losses

– Acknowledgments (ACKs), sequence numbers, timers